

ThmSmith Stage -1 proposal:
pid_dose_response_iv / v1
*Adjacent-rank copula tubes for continuous-IV derivative-sign partial
 identification*

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Locked parameters. This is a partial-identification anchor. The locked tuple is

$$(\text{pid_dose_response_iv}, \text{v1}, P, \theta, \mathcal{R}_{\Gamma, \rho, \eta}^{\text{iv-tube}}, B_{\Gamma, \rho, \eta}^{\text{slope}}).$$

Here P is the observable law of (Y, A, Z, X) , with bounded outcome $Y \in [0, 1]$, continuous dose $A \in [a_0, a_2]$, continuous instrument $Z \in \mathcal{Z} \subset \mathbb{R}$, and observed covariates X . The parameter is the adjacent-dose rank-slice finite-difference

$$\theta_{\tau, h} = \frac{\mathbb{E}\{Y(a_1) - Y(a_0) \mid R_{01} \in \tau\}}{h}, \quad a_1 = a_0 + h,$$

where $R_{01} := U_{a_0}$ is the baseline adjacent-dose rank coordinate, $U_a = F_{a, X}^-\{Y(a)\} + V_a(F_{a, X}\{Y(a)\} - F_{a, X}^-\{Y(a)\})$ is the randomized conditional probability-integral-transform rank with an independent tie-breaker $V_a \sim \text{Unif}(0, 1)$, and $\tau \subset [0, 1]$ is a fixed interval with positive length. Equivalently, when $Y(a) \mid X$ is continuous, $U_a = F_{a, X}\{Y(a)\}$. The identifying restrictions $\mathcal{R}_{\Gamma, \rho, \eta}^{\text{iv-tube}}$ are consistency, positivity for A and Z , well-defined randomized ranks, exclusion and conditional independence of Z from the latent potential-outcome array, a continuous-treatment marginal sensitivity bound Γ on dose-specific hidden-confounding tilts, a finite vector $m(Y(a_j), X) \in \mathbb{R}^K$ of IV moment-balance restrictions with tolerance η , and an adjacent-rank copula tube requiring the copula of (U_{a_0}, U_{a_1}) within X -strata to lie within sup-norm distance ρ of the comonotone copula on $\tau \times [0, 1]$. Thus the target conditions on the explicitly pinned coordinate $R_{01} = U_{a_0}$, while the copula $C_{01, x}$ controls how the adjacent rank U_{a_1} can

be paired with it. The bounding functional is the sharp interval

$$B_{\Gamma, \rho, \eta}^{\text{slope}}(P; \tau, h) = \left[\inf_{Q \in \mathcal{M}(P, \mathcal{R}_{\Gamma, \rho, \eta}^{\text{iv-tube}})} \theta_{\tau, h}(Q), \sup_{Q \in \mathcal{M}(P, \mathcal{R}_{\Gamma, \rho, \eta}^{\text{iv-tube}})} \theta_{\tau, h}(Q) \right],$$

where $\mathcal{M}(P, \mathcal{R}_{\Gamma, \rho, \eta}^{\text{iv-tube}})$ is the set of full-data laws with observed margin P satisfying the stated restrictions.

Abstract

Continuous-dose sensitivity analyses now give sharp pointwise bands for average potential outcomes under hidden confounding, but local scientific questions often concern the sign of a marginal dose-response derivative. This proposal asks whether a continuous instrument plus a bounded adjacent-rank copula tube can certify the sign of a baseline-rank-slice finite difference even when the two one-dose rank-slice marginal bands overlap enough to leave the ordinary marginal slope sign ambiguous. The routine theorem is a finite sharp-bounds representation: after discretizing the adjacent rank slice and the instrument support, the identified slope interval is the projection of one joint coupling problem rather than a product of dosewise rank-slice intervals. The flagship conjecture is that the adjacent-rank copula creates a strict sign phase transition: for small enough copula radius ρ and informative enough IV moment tolerance η , the sharp lower slope bound is positive although the one-dose rank-slice slope interval straddles zero. Resolving this would turn rank-dependence information into an auditable local derivative-sign certificate for continuous treatments under unmeasured confounding.

1 Introduction

Why the problem is important. Many continuous-treatment decisions are local: whether one more unit of exposure helps or harms a clinically relevant subgroup, whether a policy subsidy has a positive marginal return near the current level, or whether a regulator can certify that a small exposure increase is damaging. Pointwise sensitivity bands for the dose-response curve are useful, but they can be inconclusive for derivatives because two dosewise adversaries can be chosen independently. If adjacent potential-outcome ranks cannot be paired arbitrarily, and if an instrument supplies extra marginal-distribution information, the local derivative sign may be more identifiable than the dosewise mean envelope suggests.

The main finding (proposed). Theorem 1 states the bookkeeping result: for fixed adjacent doses, rank bins, instrument bins, and IV moment functions, sharp partial identification of the baseline-rank-slice slope is a finite linear-programming projection over one joint coupling of $Y(a_0)$, $Y(a_1)$, latent confounding tilts, and IV-compatible margins. Conjecture 1 is the novelty kernel: there are observable laws for which the one-dose rank-slice sensitivity intervals imply a finite-difference interval containing zero, while the adjacent-rank copula tube and IV moment restrictions force the sharp baseline-rank-slice finite-difference interval to be strictly positive. Conjecture 2 records the expected boundary form: the lower slope endpoint is monotone nondecreasing as the copula tube tightens, and crosses zero at a finite critical radius in a two-rank-bin witness.

What is new. The closest published comparator is [Baitairian et al. \(2025\)](#), which establishes sharp APO bounds under a continuous sensitivity model, together with continuous-dose hidden-confounding methods such as [Jesson et al. \(2022\)](#), [Marmarelis et al. \(2023\)](#), and [Dalal and Tchetgen Tchetgen \(2025\)](#). [Chernozhukov et al. \(2024\)](#) is the closest IV-plus-copula comparator: it uses local copula invariance to point-identify average and quantile effects for discrete and continuous treatments with a binary instrument. Those papers do not study a partially identified target whose value depends on the joint law of adjacent potential-outcome ranks after conditioning on the pinned baseline rank $R_{01} = U_{a_0}$. The proposed gap is not a generic shape restriction: it is the incompatibility of independently sharp one-dose rank-slice adversaries with a single adjacent-rank copula tube and IV-compatible moment margins.

How it positions in the literature. This is a partial-identification proposal in the continuous-treatment sensitivity, IV, and copula-bounds cluster. It uses the continuous-IV literature as marginal information about potential-outcome distributions, and the treatment-effect-distribution literature as the warning that pointwise sharp bounds need not be uniformly or jointly sharp. It deliberately avoids the panel / linear-projection cluster: the object is a sharp identified interval for a continuous-dose rank-slice derivative, not a regression coefficient.

2 Related work

[Baitairian et al. \(2025\)](#) is the closest continuous-sensitivity comparator because it gives sharp bounds for the average potential outcome under a continuous sensitivity model; this proposal changes the target from a point-wise APO to a baseline-rank finite-difference requiring one cross-dose copula. [Jesson et al. \(2022\)](#) introduces the continuous-treatment marginal sensitivity model and scalable bounds for continuous-valued interventions; the present proposal borrows the hidden-confounding radius idea but adds IV moment margins and rank-copula dependence restrictions. [Marmarelis et al. \(2023\)](#) develops partial identification for average and conditional average continuous-dose effects under hidden confounding; the gap is that their dose-response envelope does not characterize rank-slice slopes. [Dalal and Tchetgen Tchetgen \(2025\)](#) extends Rosenbaum and marginal sensitivity frameworks to endogenous continuous exposures with dose-varying sensitivity functions; the proposed result asks whether a local sign can be certified by rank compatibility even when their global curve envelope remains sign-ambiguous.

[Kasy \(2014\)](#) studies continuous IV triangular systems with unrestricted heterogeneity and states a rich-support point-identification theorem plus partial identification under insufficient support, but the corrigendum by [Hoderlein et al. \(2017\)](#) explains that the central theorem needs the stronger one-dimensional monotone first-stage heterogeneity condition used by [Imbens and Newey \(2009\)](#); this proposal works in the insufficient-support / hidden-confounding regime and asks for a derivative-sign partial-ID certificate rather than relying on that point-ID claim. [Torgovitsky \(2015\)](#) shows that scalar heterogeneity restrictions can point-identify nonseparable continuous-treatment models with small-support instruments; the contrast here is that no scalar heterogeneity point-identification assumption is imposed. [D’Haultfoeulle and Fevrier \(2015\)](#) provides a discrete-instrument triangular point-identification route under monotonicity and support/intersection conditions; the present proposal treats failures of such point-ID conditions as a partial-ID problem with copula dependence restrictions. [Chernozhukov and Hansen \(2005\)](#) is the quantile-IV benchmark; it motivates rank-sensitive IV targets, while this proposal bounds a rank-slice potential-outcome derivative rather than point-identifying a structural quantile function. [Chernozhukov et al. \(2024\)](#) is especially close because it combines binary instruments with a local copula invariance restriction to identify average and quantile treatment effects for discrete and continuous treatments; the distinction is that this proposal keeps the continuous-IV sensitivity model par-

tially identified and studies an adjacent-rank slope sign under a copula tube, not point identification under invariance. Average-derivative estimands for continuous treatments motivate the local sign target, but the object here is a bounded partial-ID finite difference inside a baseline rank slice, not an average derivative point estimand.

[Firpo and Ridder \(2019\)](#) proves that pointwise Makarov bounds for treatment-effect distribution objects are not uniformly sharp; this is the main conceptual warning behind the proposal’s claim that dosewise sharp APO intervals may be jointly nonsharp for slopes. [Fan et al. \(2017\)](#) studies partial identification of functionals of the joint distribution of potential outcomes and shows how copula information can matter for non-ATE targets; the present proposal moves that logic to adjacent continuous doses and an IV/sensitivity setting. [Dorn and Guo \(2023\)](#) develops sharp marginal-sensitivity bounds via quantile balancing, relevant because it shows how quantile restrictions can sharpen sensitivity analysis; this proposal uses a different quantile object, namely an adjacent-dose rank-copula tube.

The in-catalogue `ThmSmith` prior art is thin for this anchor. `doc/API.md` section 3 records `ThmSmith/PartialID/` as an empty seed area for Manski-family bounds, IV bounds, sensitivity analysis, and shape restrictions, so the live catalogue does not already contain a continuous-dose IV copula-slope theorem. The closest banked proposal is `pid_dose_response_shape`: shape-closed sensitivity bounds for monotone-concave continuous dose response; its kernel is finite-grid monotone-concave curve closure without an IV and without a rank-slice derivative. The present kernel is therefore not a relabelling of that banked angle: it replaces shape closure of marginal means by joint sharpness and sign certification for an adjacent-rank slope under IV-compatible margins.

3 Formal setup

Let P denote the observable law of (Y, A, Z, X) . Potential outcomes are $\{Y(a) : a \in [a_0, a_2]\}$, with $Y = Y(A)$ under consistency. The two adjacent design doses are $a_0 < a_1$, $h = a_1 - a_0$, and $\tau \subset [0, 1]$ is a fixed rank slice. For any full-data law Q , let $F_{a,x}$ be the conditional distribution function of $Y(a) \mid X = x$, let $F_{a,x}^-(y) = \mathbb{P}_Q(Y(a) < y \mid X = x)$, and introduce tie-breakers $V_a \sim \text{Unif}(0, 1)$ independent conditional on $\{Y(a), X\}$. The randomized conditional rank is

$$U_a = F_{a,X}^-(Y(a)) + V_a\{F_{a,X}(Y(a)) - F_{a,X}^-(Y(a))\},$$

so $U_a \mid X \sim \text{Unif}(0, 1)$ even when $Y(a) \mid X$ has atoms; when the conditional margin is continuous this agrees with $F_{a,X}\{Y(a)\}$. Set the target coordinate to $R_{01} := U_{a_0}$, and let $C_{01,x}$ be the conditional copula of $(U_{a_0}, U_{a_1}) \mid X = x$. For $j = 0, 1$, define the one-dose rank-slice marginal endpoints

$$L_j^\tau(P, \Gamma, \eta) = \inf_Q \mathbb{E}_Q\{Y(a_j) \mid R_{01} \in \tau\}, \quad U_j^\tau(P, \Gamma, \eta) = \sup_Q \mathbb{E}_Q\{Y(a_j) \mid R_{01} \in \tau\},$$

where the infimum and supremum range over the corresponding one-dose finite-grid sensitivity model with the same observed margin P , the same baseline-rank slice R_{01} , the Γ -tilt bound, and the IV moment tolerance η , but without a cross-dose copula-tube constraint. The rank-slice finite difference is

$$\theta_{\tau,h}(Q) = \frac{\mathbb{E}_Q[Y(a_1) - Y(a_0) \mid R_{01} \in \tau]}{h}.$$

The identified set is

$$\Theta_I(P) = \{\theta_{\tau,h}(Q) : Q \in \mathcal{M}(P, \mathcal{R}_{\Gamma,\rho,\eta}^{\text{iv-tube}})\}.$$

The bounding functional $B_{\Gamma,\rho,\eta}^{\text{slope}}(P; \tau, h)$ is the interval hull of $\Theta_I(P)$.

qid=pid_dose_response_iv

specialization=v1

observable distribution P=law(Y,A,Z,X), Y in [0,1], A in [a0,a2], continuous Z

parameter theta=baseline-rank-slice adjacent finite difference theta_{\tau,h},

with R_{01}:=U_{a0} defined by the randomized conditional PIT rank

restrictions R=consistency, positivity, randomized rank well-definedness,

IV exclusion/conditional independence,

CMSM Gamma, finite IV moment tolerance eta for m(Y(a_j),X) in R^K,

adjacent-rank copula tube radius rho for C_{01,x} on tau x [0,1]

bounding functional B=sharp lower/upper envelope over full-data laws with observed ma

4 Assumptions

Assumption 1 (Bounded adjacent-dose setup). The outcome satisfies $Y(a) \in [0, 1]$ for $a \in \{a_0, a_1\}$, the observed dose support contains neighborhoods of a_0 and a_1 , $h = a_1 - a_0 > 0$, and the rank slice $\tau = [\tau_L, \tau_U] \subset (0, 1)$ has $\tau_L < \tau_U$.

Assumption 2 (Randomized conditional ranks). For each adjacent dose $a \in \{a_0, a_1\}$, the full-data law includes an auxiliary tie-breaker $V_a \sim \text{Unif}(0, 1)$, independent conditional on $\{Y(a), X\}$, and defines

$$U_a = F_{a,X}^-(Y(a)) + V_a\{F_{a,X}(Y(a)) - F_{a,X}^-(Y(a))\}.$$

The target event $R_{01} \in \tau$ means $U_{a_0} \in \tau$, and the conditional copula $C_{01,x}$ is the copula of the randomized rank pair $(U_{a_0}, U_{a_1}) \mid X = x$. If the conditional outcome margins are continuous, this assumption reduces to the ordinary probability-integral-transform ranks.

Assumption 3 (Consistency and positivity). The observed outcome is $Y = Y(A)$. Conditional treatment and instrument densities are positive on the dose and instrument bins used in the finite approximation, conditional on X .

Assumption 4 (IV margin restriction). The instrument Z is excluded from the potential-outcome equation and is independent of the latent potential-outcome array conditional on X . Fix a known finite moment vector $m(Y(a_j), X) \in \mathbb{R}^K$. In the finite relaxation, IV compatibility is encoded by the linear inequalities

$$|\mathbb{E}_Q[m_k(Y(a_j), X) \mid Z \in z_b] - \mathbb{E}_Q[m_k(Y(a_j), X)]| \leq \eta$$

for every moment coordinate k , adjacent dose $j = 0, 1$, and instrument bin z_b .

Assumption 5 (Continuous-treatment sensitivity radius). For each adjacent dose a_j , hidden confounding is bounded by a continuous marginal sensitivity radius Γ : the complete-data density tilt that maps the observed conditional law near $A = a_j$ to the potential-outcome margin for $Y(a_j)$ lies in a known closed interval of likelihood ratios with ratio at most Γ .

Assumption 6 (Adjacent-rank copula tube). Within each X -stratum, $R_{01} := U_{a_0}$ is the randomized rank coordinate from Assumption 2 that defines the target slice. The copula $C_{01,x}$ of (U_{a_0}, U_{a_1}) satisfies

$$\sup_{(u,v) \in \tau \times [0,1]} |C_{01,x}(u, v) - \min(u, v)| \leq \rho.$$

Thus conditioning in $\theta_{\tau,h}$ is always conditioning on $U_{a_0} \in \tau$, while the tube restricts the conditional distribution of the adjacent rank U_{a_1} paired with that baseline rank.

Assumption 7 (Finite grid regularity). For the finite theorem, X , Z , $Y(a_0)$, $Y(a_1)$, U_{a_0} , and U_{a_1} are represented on fixed finite partitions, the moment vector m has finite dimension K , and the feasible set defined by Assumptions 1–6 is nonempty and closed.

Assumption 8 (Two-bin sign-witness regularity). For Conjecture 1 and Conjecture 2, the finite approximation has two baseline-rank bins inside τ , two instrument bins, nondegenerate IV-induced moment variation, and one-dose rank-slice marginal sensitivity intervals whose finite-difference interval contains zero.

Assumption 9 (Observed-confounder rank identification corner). For the unconfounded sanity corner, the observed-confounder law identifies $F_{a_0,x}$, $F_{a_1,x}$, the randomized rank construction in Assumption 2, and the adjacent copula $C_{01,x}$, and this identified copula satisfies Assumption 6.

5 Main results

Theorem 1 (Theorem 1: finite sharp coupling projection). *Under Assumptions 1–7, the finite-grid sharp identified set for $\theta_{\tau,h}$ is the projection of a nonempty compact polytope whose variables are the joint cell probabilities of $(Y(a_0), Y(a_1), U_{a_0}, U_{a_1}, Z, X)$ and the dose-specific sensitivity tilts. The linear IV moment inequalities in Assumption 4 and the copula-tube inequalities in Assumption 6 are part of the finite constraint system defining this polytope. The lower and upper endpoints of $B_{\Gamma,\rho,\eta}^{\text{slope}}(P; \tau, h)$ are attained by linear programs over this polytope.*

Why this is new and worth studying. This theorem is not the field-level novelty claim: it is the finite sharpness object needed before testing the conjectured sign phenomenon. Its closest prior art is the pointwise sharp APO construction in Baitairian et al. (2025) and the finite-dimensional joint-distribution bounds in Firpo and Ridder (2019) and Fan et al. (2017); the difference is that the variables include one adjacent-dose rank coupling plus IV-compatible potential-outcome margins. Formalizing this projection would give the repository a clean partial-ID target for continuous-dose IV sensitivity problems.

Conjecture 1 (Conjecture 1: adjacent-rank sign certificate (NOT YET PROVED)). *Under Assumptions 1–8, there exist observable laws P , sensitivity radii $\Gamma > 1$, IV moment tolerances $\eta > 0$, and copula radii $\rho > 0$ such that the one-dose baseline-rank-slice marginal sensitivity interval*

$$\left[\frac{L_1^\tau(P, \Gamma, \eta) - U_0^\tau(P, \Gamma, \eta)}{h}, \frac{U_1^\tau(P, \Gamma, \eta) - L_0^\tau(P, \Gamma, \eta)}{h} \right]$$

contains zero, while the sharp rank-slice interval $B_{\Gamma,\rho,\eta}^{\text{slope}}(P; \tau, h)$ is strictly positive.

Why this is new and worth studying. The closest sensitivity paper is [Baitairian et al. \(2025\)](#): their APO bounds can be sharp at each dose, but they do not ask whether two independently sharp dose adversaries can be pasted through one adjacent-rank copula. The closest IV-copula paper is [Chernozhukov et al. \(2024\)](#), which uses local copula invariance for point identification with a binary instrument; it does not give a partial-ID sign certificate for an adjacent baseline-rank slope under a sensitivity radius and copula tube. Resolving this conjecture would let analysts report a local derivative-sign certificate under hidden confounding, instead of stopping at an inconclusive difference of pointwise bands.

Conjecture 2 (Conjecture 2: two-bin copula-radius phase transition (NOT YET PROVED)). *In the two-rank-bin witness of Assumption 8, holding P , Γ , the IV moment vector m , and η fixed, the lower endpoint*

$$\underline{\theta}(\rho) = \inf B_{\Gamma, \rho, \eta}^{\text{slope}}(P; \tau, h)$$

is nonincreasing in ρ . Moreover, for some nonempty open set of witness laws there is a critical radius $\rho^ \in (0, 1)$ such that $\underline{\theta}(\rho) > 0$ for $\rho < \rho^*$, $\underline{\theta}(\rho) = 0$ at $\rho = \rho^*$, and the one-dose baseline-rank-slice marginal finite-difference interval remains sign-ambiguous throughout.*

Why this is new and worth studying. [Firpo and Ridder \(2019\)](#) shows that pointwise sharp bounds can fail to be jointly sharp for process functionals; this conjecture proposes a concrete continuous-dose analogue where the boundary is governed by a copula-radius parameter. The gap to standard copula-bounds work such as [Fan et al. \(2017\)](#) and IV local-copula work such as [Chernozhukov et al. \(2024\)](#) is the sensitivity-margin plus copula-radius phase boundary for a local derivative target. A phase-transition theorem would give practitioners a transparent robustness curve: how close adjacent ranks must be before a derivative sign becomes certified.

Theorem 2 (Theorem 2: no-tube reduction to one-dose rank-slice bounds). *Under Assumptions 1–7, if $\rho = 1$ imposes no effective restriction on the adjacent copula and η is set large enough that the IV moment inequalities in Assumption 4 are vacuous, then $B_{\Gamma, \rho, \eta}^{\text{slope}}(P; \tau, h)$ contains the one-dose rank-slice finite-difference interval*

$$\left[\frac{L_1^\tau(P, \Gamma, \eta) - U_0^\tau(P, \Gamma, \eta)}{h}, \frac{U_1^\tau(P, \Gamma, \eta) - L_0^\tau(P, \Gamma, \eta)}{h} \right],$$

where L_j^τ, U_j^τ are the rank-slice marginal endpoints defined in Section 6.

Why this is new and worth studying. This is a sanity theorem rather than the novelty kernel. It checks that the proposed object does not create derivative-sign information by notation alone: when the cross-dose copula and IV moment restrictions are removed, the proposal collapses back to the product of one-dose rank-slice sensitivity envelopes, the rank-slice analogue of the pointwise continuous-sensitivity bounds studied by [Jesson et al. \(2022\)](#), [Marmarelis et al. \(2023\)](#), and [Baitairian et al. \(2025\)](#).

6 Sanity-check corollaries

Corollary 1 (Unconfounded adjacent-rank collapse). *Under Assumptions 1–6 and Assumption 9, if $\Gamma = 1$ and $\eta = 0$, then $B_{1,\rho,0}^{\text{slope}}(P; \tau, h)$ is the singleton containing the corresponding observed-confounder rank-slice finite difference.*

Under Assumptions 1–6 and Assumption 9, the statement collapses cleanly because $\Gamma = 1$ removes hidden-confounding tilts, $\eta = 0$ makes the finite IV moment margins exact, and the known adjacent copula fixes the joint mean difference conditional on the pinned event $R_{01} = U_{a_0} \in \tau$.

Corollary 2 (No shared-rank information gives no sign certificate). *Under Assumptions 1–7, if $\rho = 1$, η is large enough to make Assumption 4 vacuous, and the one-dose rank-slice finite-difference interval in Theorem 2 contains zero, then the proposed rank-slice analysis cannot certify a strictly positive derivative sign.*

Under Assumptions 1–7, this is the direct sanity check for Theorem 2: without the copula tube or finite IV moment restrictions, independent one-dose rank-slice endpoint choices remain feasible in the product relaxation, so a positive sign cannot be forced when that product interval contains zero.

7 Proof-sketch route

The mathematical route is finite-dimensional convex analysis plus a two-bin witness calculation. Theorem 1 follows by writing the observed margins, randomized-rank bin masses, sensitivity-ratio bounds, finite IV moment inequalities, copula-tube inequalities, and probability normalization constraints as linear equalities and inequalities on a finite joint table; compactness gives attained extrema. Theorem 2 is proved by constructing product couplings that attain the one-dose rank-slice marginal endpoints when

the cross-dose and IV moment constraints are vacuous. The conjectural work is the explicit two-baseline-rank-bin algebra: write a 2×2 table p_{rs} for baseline rank bin $r \in \{L, H\}$ and adjacent rank bin $s \in \{L, H\}$, impose fixed row and column margins, impose $|p_{LL} - (p_L \wedge p_L)| \leq \rho$ as the local copula-radius inequality, and add one signed IV moment inequality that prevents the low-slope cell from absorbing all baseline-low mass. Solving that four-cell linear program should expose the threshold at which the lower finite difference crosses zero.

8 Honest scope flags

The finite LP representation and the no-tube reduction are labelled as theorems because they are routine once the finite partitions and restrictions are fixed. The sign certificate and the critical-radius phase transition are labelled as conjectures because the two-bin witness has not yet been algebraically enumerated. The copula-tube restriction is deliberately local to adjacent doses and a baseline rank slice $R_{01} = U_{a_0} \in \tau$; the proposal does not claim point identification of the full dose-response curve or the full treatment-effect distribution. The IV tolerance η is now fixed as a finite vector of linear moment inequalities; variants based on full conditional distributions or Wasserstein distance are outside this locked proposal.

9 Iteration trail

- v1 (pivot) — initial draft for the adjacent-rank copula-tube angle after the dynamic-IV compliance seed was burned; prior verdict: none.
- v2 (revise) — pinned $R_{01} = U_{a_0}$, defined one-dose rank-slice marginal endpoints, fixed η as finite IV moment inequalities, and repaired the no-tube sanity reduction; prior verdict: REVISE.
- v3 (revise) — replaced ordinary ranks by randomized conditional ranks to make copulas well-defined under atoms, narrowed the Kasy comparison with the 2017 corrigendum, and added a two-by-two witness-table route; prior verdict: REVISE.
- v4 (revise) — added the local-copula binary-IV comparator of Chernozhukov, Fernandez-Val, Han, and Wuthrich (2024) and distinguished point identification under copula invariance from the rank-slice partial-ID slope kernel; prior verdict: REVISE.

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